Lyondell South Asia Pte. Ltd.

78 Shenton Way #25-01,
Singapore 079120 Singapore

Certificate Of Analysis

Certificate of Analysis Contact:

Ship-To Address:

VIETNAM

Lyondell Material : 8820 Propylene Glycol USP/EP
Batch Number : GX250204L1
Vehicle Number : YM INITIATIVE 335N
Estimated Quantity : 17,200 MT

VIETNAM

Customer Order No. : 2398 OP
Customer Number : 270483
Date Shipped : February 13, 2025 :
Lyondell Order No. 5012900077 000020 :
Delivery Item No. 8108576240 000020

Test Description	Test Result	Specifications Minimum	Specifications Maximum	Unit of Measure
Vehicle Type	SCREATIVTY 270125			
Vehicle ID	SCREATIVTY270125			
Miscellaneous Text	4900821079/FDA6980-226DRUMS			
Manufacturing Date	20241124			
Expiration Date	20261124			
Acidity as acetic acid HOC 0210	< 0,0008		0,0020	weight %
Acidity (ml 0.1N NaOH/10 ml) HOC 0210	0,01		0,03	.
Appearance HOC 0650	Pass	Pass		
Assay as PG, on a dry basis HOC 0540	99,95	99,90		weight %
PG Retention Time, ID Test C HOC 0540	Pass	Pass		
Chlorides HOC 0221	< 0,5		0,5	ppm
Color, Pt-Co HOC 0641	3		10	APHA Color
Heavy Metals as Pb HOC 0274	< 1		5	weight ppm
Lead (Pb)	< 1,0		1,0	weight ppm

(*** Continued ***)

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Iron HOC 0290	< 0,10		0,50	weight ppm
Odor HOC 0610	Pass	Pass		
Suspended matter HOC 0171	Pass	Pass		
Residual Solvents USP 467	Pass	Pass		
Water HOC 0316	0,015		0,200	weight %
Ethylene Glycol	< 0,001		0,008	weight %
Diethylene Glycol	< 0,001		0,008	weight %
TYPICAL VALUES	SHOWN BELOW			
Arsenic as As HOC 0242	< 1		1	weight ppm
CONEG Metals D3335	Pass	Pass		
IR Spectrum HOC 0411	Pass	Pass		
Distillation IBP @760 mmHg HOC 0154	187,3	186,0		degrees C
Distillation DP @760 mmHg HOC 0154	188,2		189,0	degrees C
Oxidiz Subst as ml thiosulfate HOC 0204	< 0,20		0,20	Milliliter
Reducing Substances HOC 0205	Pass	Pass		
Residue on Ignition HOC 0140	0,0010		0,0050	weight %
Specific Gravity @ 20/20 ℃ HOC 0132	1,0381	1,0377	1,0389	.
Specific Gravity @ 25/25C HOC 0132	1,0356	1,0352	1,0364	.
Sulfate HOC 0225	< 0,006		0,006	weight %
Solubility @25 deg. C.	Pass	Pass		

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Test Description	Test Result	Specifications Minimum	Maximum	Unit of Measure
HOC 0160				
Refractive Index @ 20 C	1,4329	1,4310	1,4330	.
HOC 0740				

Compendia
Lyondell Chemical Company Propylene Glycol USP meets or exceeds all requirements of the United States Pharmacopeia, the European Pharmacopeia, Japanese Pharmacopeia and the Food Chemicals Codex.
This product is not for use as an Active Pharmaceutical Ingredient (API)

EU Food Regulations and Certifications
Propylene Glycol USP/EP at our European plants is HACCP certified and FSSC22000 certified. Food additive E 1520 restricted use in food.

Site of Manufacture
All propylene glycol USP sold by Lyondell Chemical Company in the United States, Canada and Central and South America is manufactured at the Lyondell Bayport plant at 10801 Choate Road, Pasadena, TX 77507.

All propylene glycol USP sold by Lyondell Chemical Company in Europe, the Middle East and Africa is manufactured at the Lyondell Chemie Nederland (RB) plant at Theemsweg 14, 3197 KM Rotterdam and Lyondell Chimie France (FL) plant at Route du Quai Mineralier, 13270, Fos-sur-Mer.
All LyondellBasell PG USP/EP batches starting with "AD" are manufactured at Fos-sur-Mer

All propylene glycol USP/EP sold by Lyondell Chemical Company in Asia is manufactured at the Lyondell Bayport plant at 10801 Choate Road, Pasadena, TX 77507 unless otherwise stated.

Date of Manufacture
For shipments from Lyondell's Bayport,Rotterdam and Fos plants the date of manufacture is shown as the Manufacturing Date in the data fields above. The date is in the format yyyymmdd, where yyyy is the year, mm is the month and dd is the day. This date of Manufacture also applies to the European packaged material shipped from our Fos facility.

For shipments from terminals in Europe, the date of manufacturing is determined from manufacturing date in the test description. The manufacturing date is show in format YYYY-MM-DD. For example, a manufacturing date of 20240211 indicates that the manufacturing date is 11th February 2024. This indicates the date that the batch was finalyzed in our facility in FOS in France.

For shipments from terminals in Europe, the batch number is shown in format Source-YY-MM-DD-sequence number. For example, a batch number of AD11100501 indicates that the batch was finalized on 05th October, 2011 at terminal ADPO.

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Test Description	Test Result	Specifications Minimum	Maximum	Unit of Measure

For shipments from the IMTT terminals in Bayonne, NJ, and Lemont, IL, the date of manufacture may be determined from the Lot Number in the Test Description portion of the Certificate of Analysis. The Lot Number is the format DATE-SOURCE-TANK. For example, a Lot Number of 50811-IMTT-14100 indicates that the material shipped from the IMTT terminal, the manufacturing date was August 11, 2005, (ymmdd) and the product shipped from tank 14100. Each bulk storage tank has a unique number.
IMTT Bayonne: 2 Commerce Street Bayonne, NJ 07002
IMTT Lemont: 13589 Main Street Lemont, IL 60439

This coding applies only to shipments made by Lyondell from Lyondell-owned or leased bulk storage tanks. It does not apply when a new lot number is assigned by a third party such as a chemical distributor.

Lot Number and Traceability
For shipments from Lyondell's plant lot traceability may be maintained through either the Batch Number or Delivery Item No. in the header or through the Inspection Lot in the data field under Test Description
Batch number is coded as Plant code-Year-month-date-sequence number (BA-Y-MM-DD-XXX).
Plant codes: RB-Botlek, FL-Fos, AD-ADPO, EC-Cogex. All LyondellBasell PG USP/EP batches starting with "EC" are manufactured at Fos-sur-Mer.

Shelf Life
The shelf life of propylene glycol USP manufactured and delivered by Lyondell Chemical Company when stored and handled under appropriate conditions is two years from the manufacturing date. Subsequent handling and/or repackaging by intermediaries such as distributors may affect shelf life. Therefore, the user of this material should confirm this shelf life via testing or with their immediate supplier.

Residual Solvents
The United States Pharmacopeia Current Edition of USP, in section <467> "Residual Solvents", specifies limits on Residual Solvents. Section <467> lists classes of Residual Solvents which may not be completely removed by practical manufacturing techniques.

None of the solvents listed as Class 1, Class 2, Class 3 or those listed in Table 4 are used as raw materials or processing aids, none are expected to be by-products of the manufacturing process, and none are added to the final product. Therefore this product meets USP guidelines for Residual Solvents.

Propylene glycol is manufactured by the reaction of propylene oxide and water and no other solvents are used. By virtue of the manufacturing process no residual propylene oxide is likely to be present in propylene glycol.

Dipropylene glycol, which is typically present at 0.10 - 0.20%, is not currently listed in USP current ed. <467> as a residual solvent.

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Test Description	Test Result	Specifications Minimum	Maximum	Unit of Measure

PG Retention Time (Identification Test C)
The retention time of the propylene glycol peak in the chromatogram of the sample solution corresponds to that obtained in the chromatogram of the standard solution.

Typical Properties
Analytical test parameters listed under Typical Properties are not tested on every lot of material based on our knowledge of the manufacturing process and controlled handling, shipping and storage of the material. Random, statistical sampling has shown that Lyondell Chemical Company Propylene Glycol USP, if tested, will fall within the limits indicated under Test Result.

This is a computer generated letter. No signature is required

Print Date: February 09, 2025
This information is available 24 hours a day at
www.CustomerXPRESS.com

ACLEE1

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